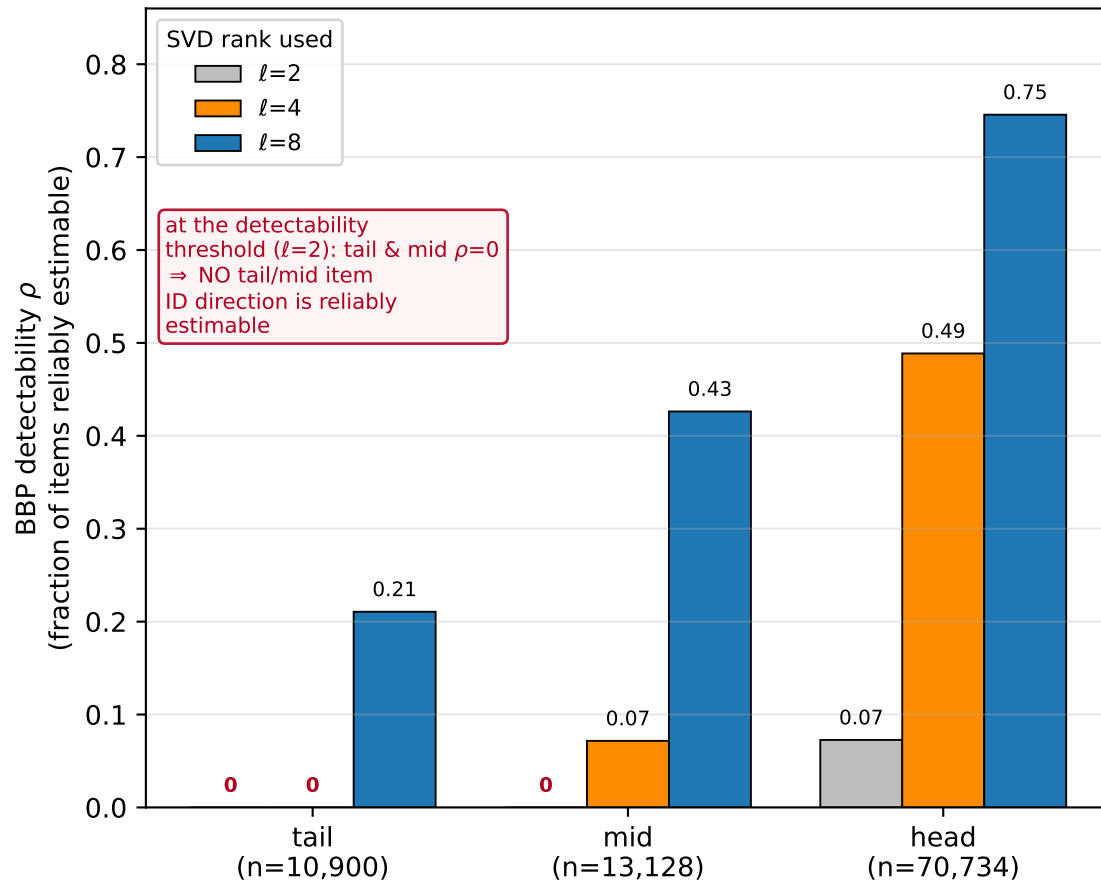
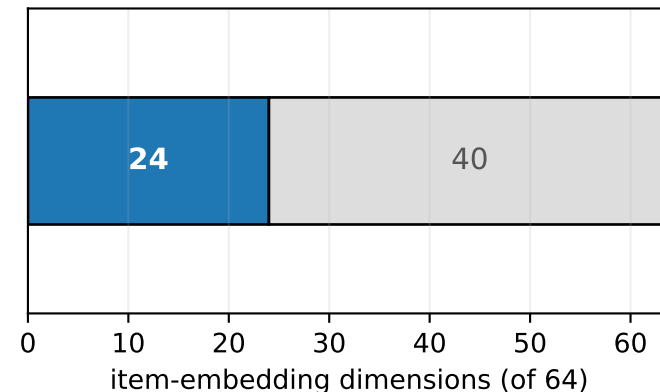


Spectral irreducibility of the long tail: why no representation-side lever rescues it

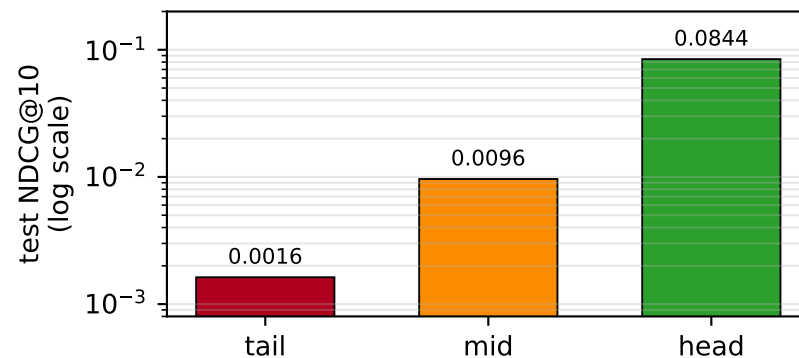
The long tail sits BELOW the BBP detectability edge (VG item ID-embedding spectrum, GD/MP diagnostic)



GD/MP hard ⇒ the item is only 24 of 64 dims
discarded (below MP edge) = 40 low-rank



**Realized accuracy tracks detectability:
tail $\approx 52\times$ below head**



Source: results_GD1_spectralshrink_VG.json (V2 stack + GD/MP spectral hard-shrink, seed 20260608, AR2023 VG 5-core LLOO, full-catalog $n_{eval}=94,762$, train-frequency terciles). BBP ρ = fraction of a tercile's items whose top- l ID singular directions clear the Marchenko-Pastur/BBP detectability edge. GD1 spectral-shrink itself HURT overall ($0.0645 < V2$ 0.0674) -- this diagnostic is the contribution that survives that negative test. BBP=Baik-Ben Arous-Peche 2005; shrinkage=Gavish-Donoho 2014.